# Principal R&D Systems Engineer - Electrophysiology

Boston Scientific · Waltham, MA, US, 02138

| | |
|---|---|
| **Job ID (this archive)** | HI-062 |
| **Employer** | Boston Scientific |
| **Location(s)** | Waltham, MA, US, 02138 |
| **Career stage (archive tag)** | Experienced |
| **Location group** | Other U.S. |
| **Majors (archive tags)** | Biomedical Engineering, Mechanical Engineering, Systems Engineering |
| **Pay listed** | $106,800 – $202,900 per year |
| **Employer requisition** | 1435062000 |
| **Recruiting status when checked** | Open when checked (Oct 10, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print. |
| **Source page** | https://jobs.bostonscientific.com/job/Waltham-Principal-R&D-Systems-Engineer-Electrophysiology-MA-02138/1435062000/ |
| **Captured** | 2026-10-07T20:13:06.542796 UTC |
| **Course sheet refs** | HI-062 |

> Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

## Job description

Additional Location(s): N/A

Diversity - Innovation - Caring - Global Collaboration - Winning Spirit - High Performance

At Boston Scientific, we'll give you the opportunity to harness all that's within you by working in teams of diverse and high-performing employees, tackling some of the most important health industry challenges. With access to the latest tools, information and training, we'll help you in advancing your skills and career. Here, you'll be supported in progressing - whatever your ambitions.

We are looking for a seasoned engineer to join our Systems Engineering department within the Electrophysiology Division at Boston Scientific. The Principal Systems Engineer will be a Systems Engineering technical lead on complex product development projects for the OPAL Mapping System. This person will be responsible for leading the design and development of novel integrated solutions for Boston Scientific's rapidly growing and evolving EP product portfolio.

The Principal Systems Engineer will be a member of a cross-functional core team. Responsibilities of the Principal Systems Engineer include: leading the development of requirements in close collaboration with marketing and internal and external stakeholders, understanding and analyzing clinical workflows, interfacing with/presenting to medical key opinion leaders, generating systems architectural concepts in collaboration with the technical team, leading and providing guidance to an interdisciplinary team of SW/Electrical/Mechanical professionals, planning test method developments, analyzing data, implementing the solutions in hardware and software according to Boston Scientific's product development procedures.

**Your responsibilities will include:**

You will be responsible for completing system engineering activities, from early phase development to market release. Your key tasks will include:

Designing or supporting the development of systems architecture aligned with business and customer needs.

Executing multiple systems engineering activities throughout the product development life cycle.

Applying critical thinking skills to solve complex systems interaction problems.

Providing expertise in multiple roles in the product development life cycle, showcasing deep knowledge of system functionality.

Advising the project management team on key product risk/benefit trade-offs.

Influencing cross-functional peers to make timely decisions and recommendations for project success.

Representing the Systems Engineering organization as the prime technical contact for assigned features, tools, or subject matters.

Providing input for submissions to regulatory bodies and governmental agencies.

Defining interface requirements between systems and maintaining traceability of requirements documentation.

Managing changes to interfaces during development and throughout the product lifecycle.

Ensuring configuration compatibility through the product lifecycle.

**Qualifications :**

Bachelor's degree in Engineering and/or Science with 10+ years of experience in developing highly complex systems. 7+ years of experience as a Systems Engineer in the medical/diagnostics space, focusing on electromechanical systems.

Demonstrated working knowledge of systems engineering principles, requirements management, and medical device regulations.

Ability to lead teams through a structured approach to solving complex technical problems.

Experience with requirement management, collaboration, and issue tracking tools.

Experience in product concept development, customer interaction, verification of engineering requirements, and validation of customer needs/design requirements.

Practical knowledge of project management concepts and experience with Agile concepts including scrums.

Leadership skills with the capability to make trade-offs between risks, benefits, and performance.

Understanding and experience in the design, development, and testing of software based medical devices.

Familiarity with developing Test Methods and Test Equipment.

Experience with industry-based standards, including testing for safety and EMC according to 60601-1

**Preferred Qualifications**

Experience with cyber security and IEC 62304.

Usability engineering experience.

Electrophysiology knowledge.

Medical Device experience working with imaging and signal processing systems (EEG, Electrophysiology, etc.).

Experience conducting simulations, mathematical analysis and/or system interaction analysis.

At least 6 years of engineering experience in mixed HW/SW systems.

Technical experience with digital and analog low noise systems.

Strong written and verbal communication, judgment, decision-making, and critical thinking skills.

Ability to build strong relationships across the organization and with external stakeholders.

Requisition ID: 635093

Minimum Salary: $ 106800

Maximum Salary: $ 202900

The anticipated compensation listed above and the value of core and optional employee benefits offered by Boston Scientific (BSC) - see www.bscbenefitsconnect.com- will vary based on actual location of the position and other pertinent factors considered in determining actual compensation for the role. Compensation will be commensurate with demonstrable level of experience and training, pertinent education including licensure and certifications, among other relevant business or organizational needs. At BSC, it is not typical for an individual to be hired near the bottom or top of the anticipated salary range listed above.

Compensation for non-exempt (hourly), non-sales roles may also include variable compensation from time to time (e.g., any overtime and shift differential) and annual bonus target (subject to plan eligibility and other requirements).

Compensation for exempt, non-sales roles may also include variable compensation, i.e., annual bonus target and long-term incentives (subject to plan eligibility and other requirements).

For MA positions : It is unlawful to require or administer a lie detector test for employment. Violators are subject to criminal penalties and civil liability.

Boston Scientific transforms lives through innovative medical technologies that improve the health of patients around the world. As a global medical technology leader for more than 45 years, we advance science for life by providing a broad range of high-performance solutions that address unmet patient needs and reduce the cost of healthcare. Our portfolio of devices and therapies helps physicians diagnose and treat complex cardiovascular, respiratory, digestive, oncological, neurological and urological diseases and conditions. Learn more at www.bostonscientific.com and follow us on LinkedIn .

Boston Scientific Corporation has been and will continue to be an equal opportunity employer. To ensure full implementation of its equal employment policy, the Company will continue to take steps to assure that recruitment, hiring, assignment, promotion, compensation, and all other personnel decisions are made and administered without regard to race, religion, color, national origin, citizenship, sex, sexual orientation, gender identity, gender expression, veteran status, age, mental or physical disability, genetic information or any other protected class.

Please be advised that certain US based positions, including without limitation field sales and service positions that call on hospitals and/or health care centers, require acceptable proof of COVID-19 vaccination status. Candidates will be notified during the interview and selection process if the role(s) for which they have applied require proof of vaccination as a condition of employment. Boston Scientific continues to evaluate its policies and protocols regarding the COVID-19 vaccine and will comply with all applicable state and federal law and healthcare credentialing requirements. As employees of the Company, you will be expected to meet the ongoing requirements for your roles, including any new requirements, should the Company’s policies or protocols change with regard to COVID-19 vaccination.

Among other requirements, Boston Scientific maintains specific prohibited substance test requirements for safety-sensitive positions. This role is deemed safety-sensitive and, as such, candidates will be subject to a prohibited substance test as a requirement. The goal of the prohibited substance testing is to increase workplace safety in compliance with the applicable law.

## Notes for ECS 1100 students

### How this connects to your major

Biomedical Engineering: device design, clinical use, testing, quality, human factors or biomedical research. These are teaching annotations, not an employer guarantee of eligibility.

Use the degree and experience evidence to distinguish a first job from a later career step.

BME: Teaching interpretation of role tasks; major eligibility is not explicitly confirmed

### Skills mentioned in the posting

Modeling/simulation, Verification/validation, Clinical knowledge, Communication/teamwork

### Course sheet tags

Majors (teaching tags): BME; Career stage (teaching tag): Senior / leadership; Geography: Other US location; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Not stated in captured metadata.

## Source record

Original posting https://jobs.bostonscientific.com/job/Waltham-Principal-R&D-Systems-Engineer-Electrophysiology-MA-02138/1435062000/
Accessed UTC: 2026-10-07T20:13:06.542796+00:00
Saved raw source: sources/postings/Boston_Scientific_4b19c5185ece.html
SHA-256: 4b19c5185ece5848716e40eafa4d836f1e31e1552cc944b345ab8867c9ea7930